

Practical List: Tuples in Python

1. Creating Tuples

Q1: Write a Python program to create a tuple of numbers.

```
my_tuple = (1, 2, 3, 4, 5)
print(my_tuple)
```

Q2: Create a tuple with mixed data types (int, string, float).

2. Accessing Tuple Elements

Q3: Write a program to access the first and last element of a tuple.

Q4: Access elements from index 1 to 3 in a tuple.

3. Basic Tuple Operations

Q5: Concatenate two tuples and print the result.

Q6: Repeat a tuple three times.

4. Tuple Processing Functions

Q7: Find the length of a tuple.

Q8: Find the maximum and minimum value in a tuple.

5. Nested Tuples

Q9: Create a nested tuple and access an inner element.

Q10: Write a program to iterate through a nested tuple.

```
nested = ((1, 2), (3, 4), (5, 6))
for inner in nested:
    for val in inner:
        print(val)
```

6. Inserting, Modifying, and Deleting Elements in a Tuple

Q11: Show how to insert an element into a tuple using conversion to list.

Q12: Modify an element of a tuple using conversion to list.

Q13: Delete an element from a tuple using conversion to list.

Creating tuples

```
t1 = ()
t2 = (10, 20, 30, 40)
t3 = ("apple", 2, 4.5, True)
t4 = (1, (2, 3), (4, 5))

print(t1)
print(t2)
print(t3)
print(t4)

t = (5, 10, 15, 20, 25)

print("First element:", t[0])
print("Last element:", t[-1])
print("Slice (2nd to 4th):", t[1:4])

t1 = (1, 2, 3)
t2 = (4, 5, 6)

# Concatenation
print("Concatenation:", t1 + t2)

# Repetition
print("Repetition:", t1 * 3)

# Membership
print("Is 3 in t1?", 3 in t1)
print("Is 10 not in t2?", 10 not in t2)

t = (8, 3, 5, 1, 9,)

print("Length:", len(t))
print("Max:", max(t))
print("Min:", min(t))
print("Sum:", sum(t))
print("Sorted:", sorted(t))

#nested
nested = ("python", (10, 20, 30), ["a", "b", "c"])

print("Whole tuple:", nested)
print("Second element:", nested[1])
print("Second element, second item:", nested[1][1])
print("List inside tuple, first element:", nested[2][0])
```

```
t = (10, 20, 30, 40)
print("Original tuple:", t)

# Convert to list
temp = list(t)

# Insert
temp.insert(2, 25)
print("After insertion:", tuple(temp))

# Modify
temp[1] = 200
print("After modification:", tuple(temp))

# Delete
del temp[3]
print("After deletion:", tuple(temp))
```